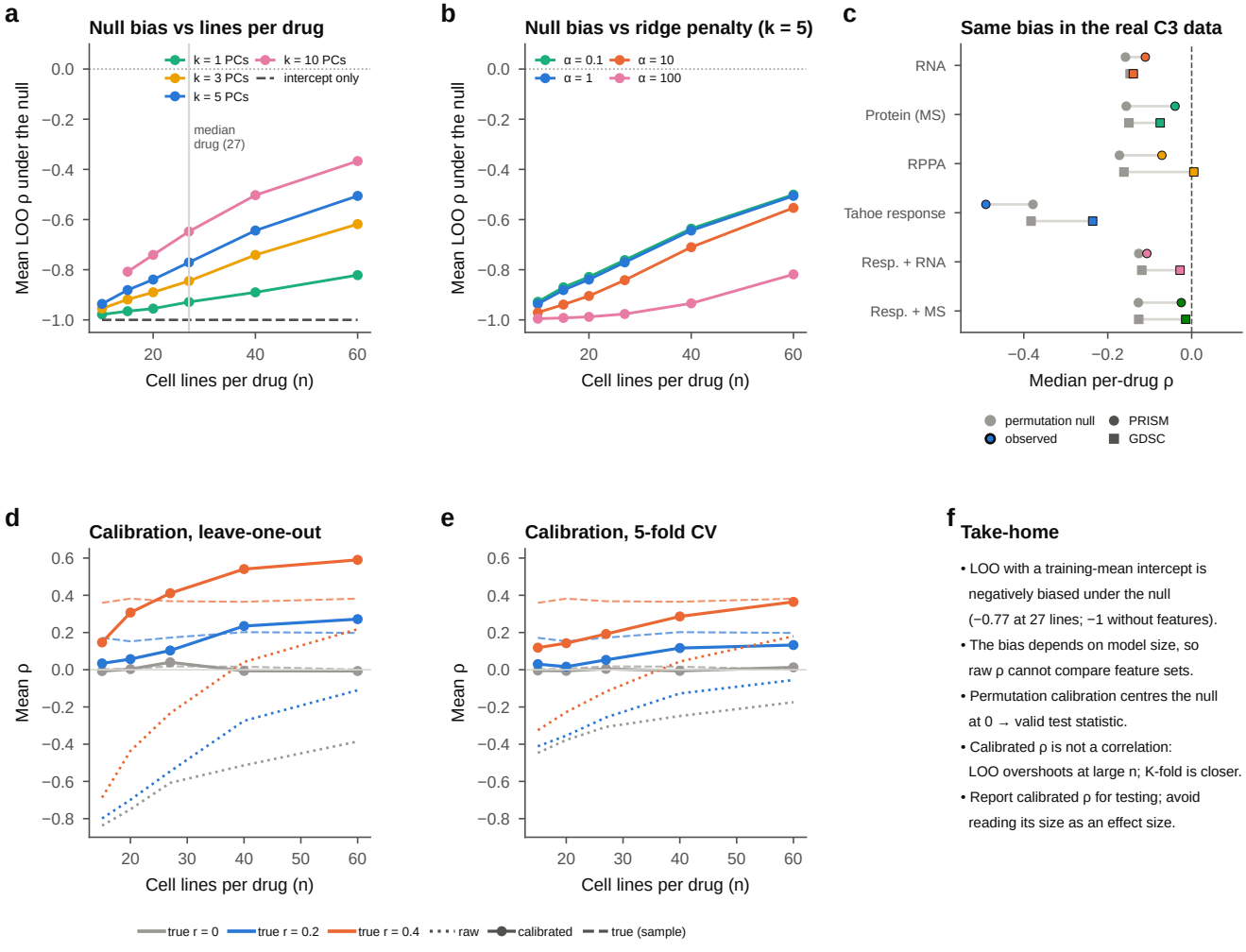
a
Null bias vs lines per drug
k = 1 PCs
k = 10 PCs
k = 3 PCs
intercept only
k = 5 PCs
median drug (27)
Mean LOO ρ under the null
Cell lines per drug (n)
b
Null bias vs ridge penalty (k = 5)
α = 0.1
α = 10
α = 1
α = 100
Mean LOO ρ under the null
Cell lines per drug (n)
c
Same bias in the real C3 data
RNA
Protein (MS)
RPPA
Tahoe response
Resp. + RNA
Resp. + MS
Median per-drug ρ
permutation null
observed
PRISM
GDSC
d
Calibration, leave-one-out
Mean ρ
Cell lines per drug (n)
e
Calibration, 5-fold CV
Mean ρ
Cell lines per drug (n)
true r = 0
true r = 0.2
true r = 0.4
raw
calibrated
true (sample)
f
Take-home
• LOO with a training-mean intercept is negatively biased under the null (−0.77 at 27 lines; −1 without features).
• The bias depends on model size, so raw ρ cannot compare feature sets.
• Permutation calibration centres the null at 0 → valid test statistic.
• Calibrated ρ is not a correlation: LOO overshoots at large n; K-fold is closer.
• Report calibrated ρ for testing; avoid reading its size as an effect size.